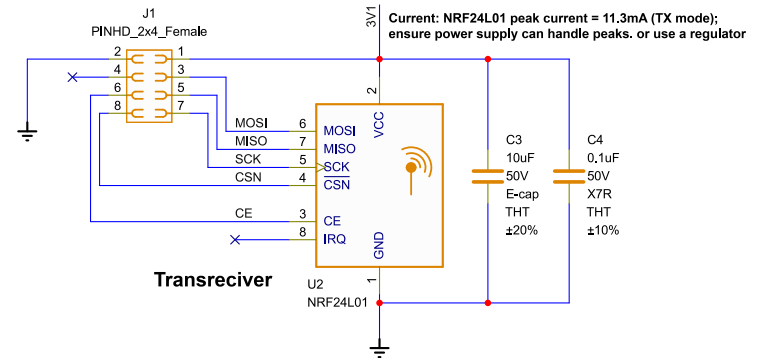
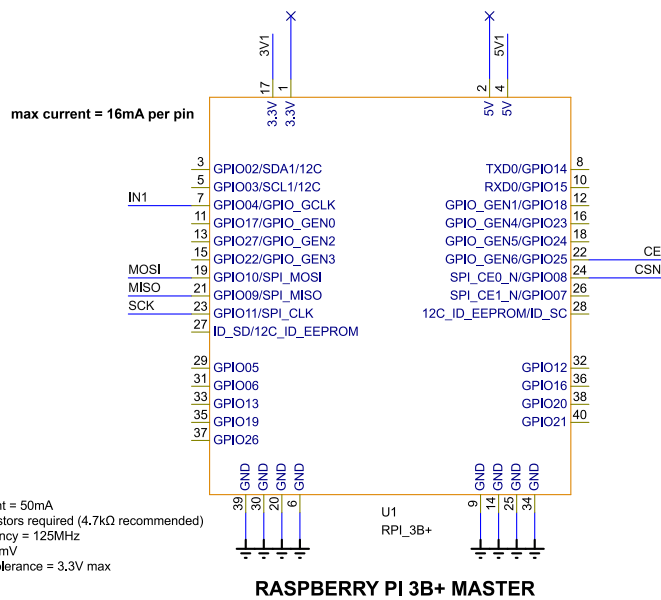
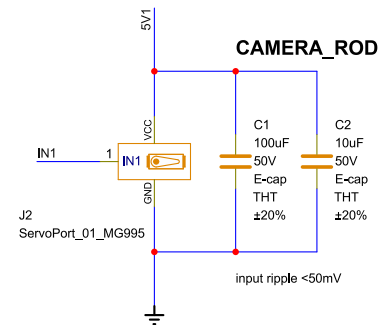




Voltage: VCC = 3.3V (±5%), NRF24L01 maximum rating = 3.6V.
SPI Lines: MOSI, MISO, SCK, and CSN require series resistors (330Ω recommended) for protection.
IRQ Pin: Pull-up resistor (10K) required if not connected to the host controller.



CL/HDBME/17/01

ICBT CAMPUS

Sheet: /Rpi_con/

File: raspberryPi_connections.kicad_sch

Title: RaspberryPi connections

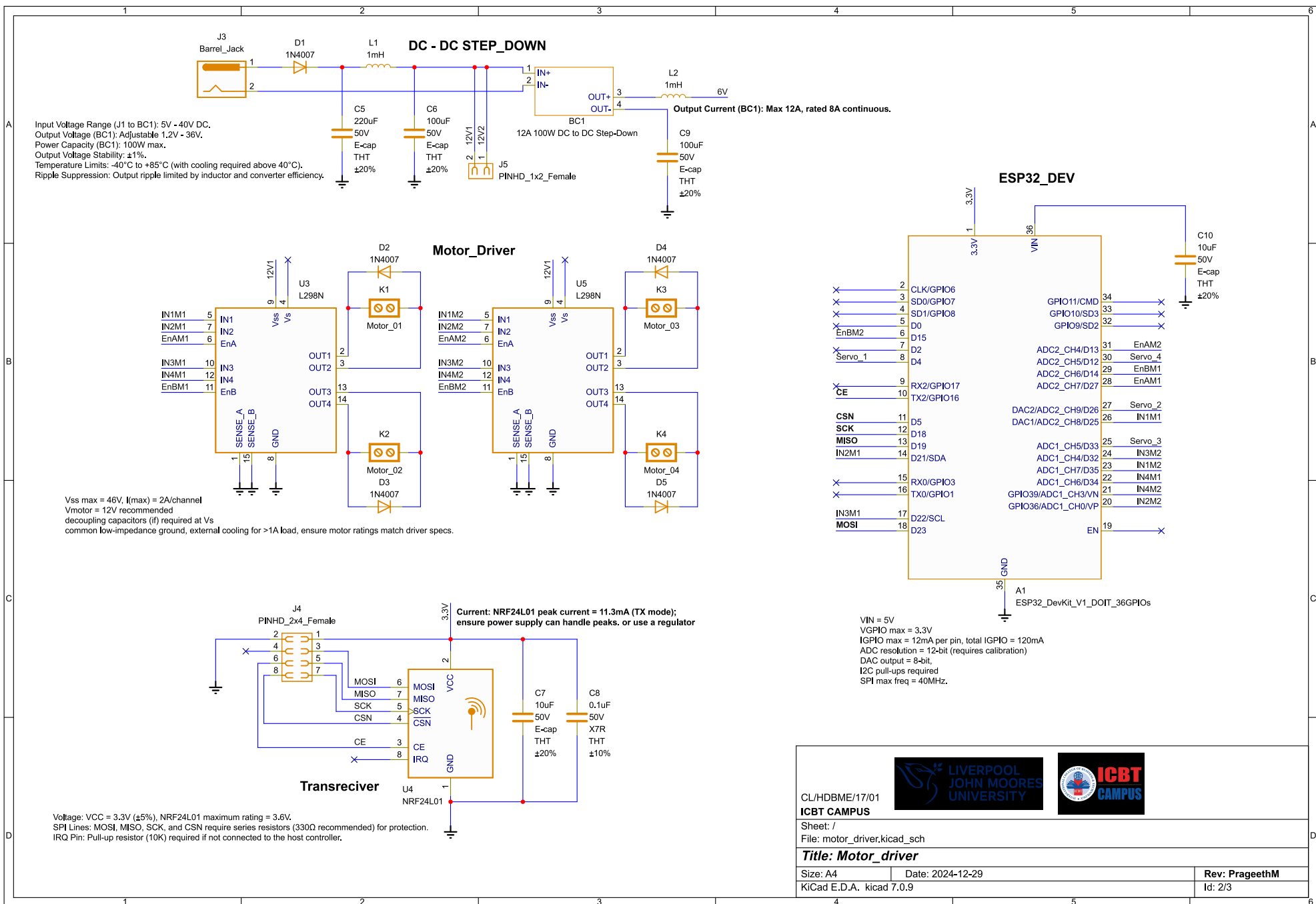
Size: A4

Date: 2024-12-30

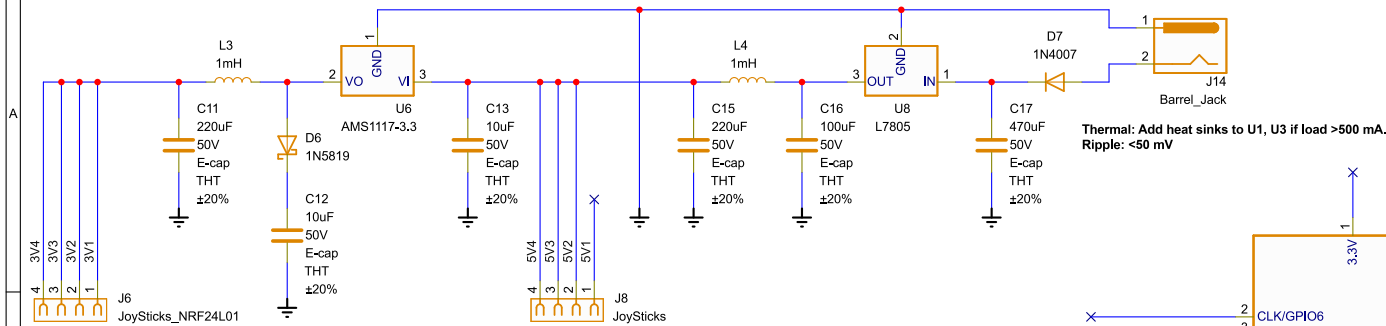
KiCad E.D.A. Kicad 7.0.9

Rev: PrageethM

Id: 1/3

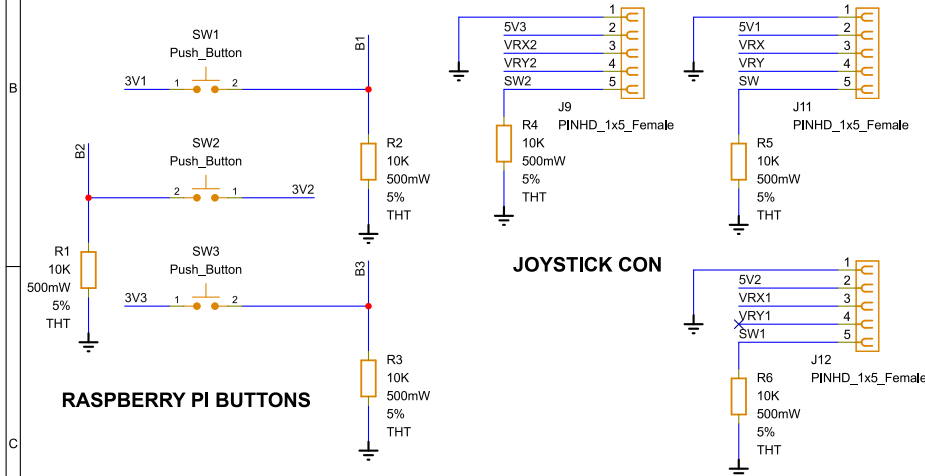


VOLTAGE REGULATOR



Thermal: Add heat sinks to U1, U3 if load >500 mA.
Ripple: <50 mV

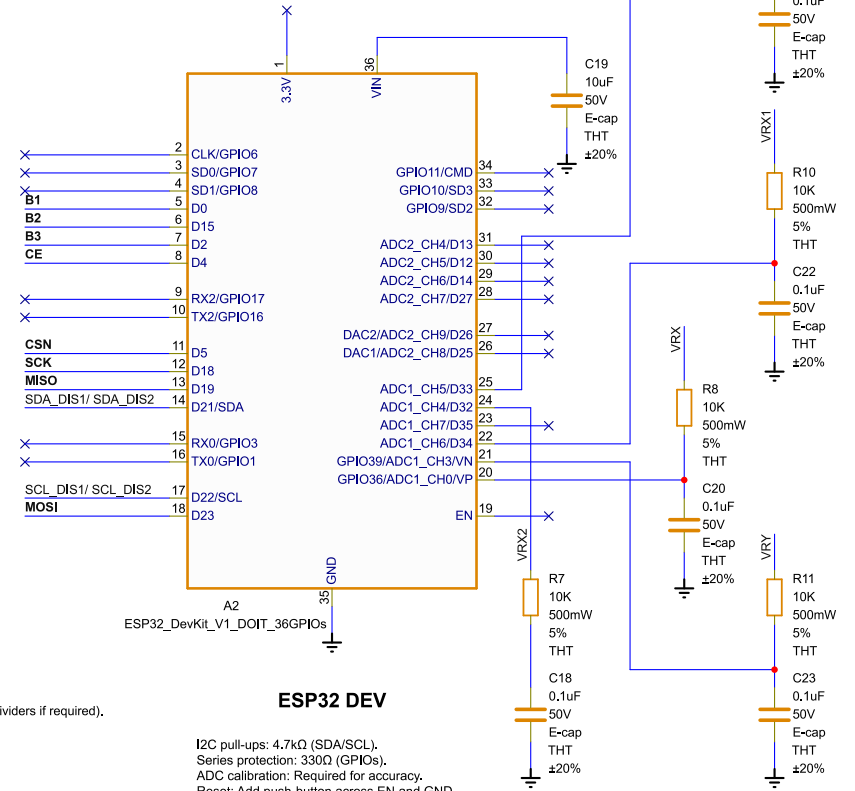
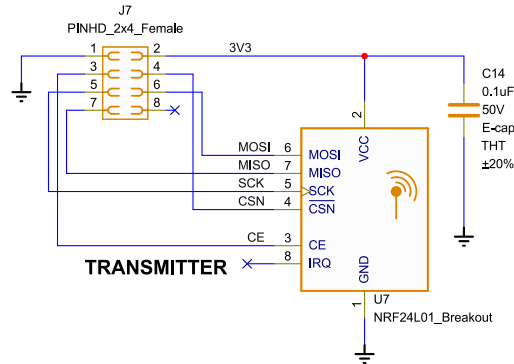
JOYSTICK CON



Voltage Levels: 5V (ensure VRX, VRY signals are within 3.3V for ESP GPIO, use voltage dividers if required).
Switch Protection: Add 330Ω series resistors on SW1, SW2 lines for current limiting.
Verify mechanical alignment for consistent VRX, VRY readings.

RASPBERRY PI BUTTONS

SPI lines (MOSI, MISO, SCK, CS): Series resistors (330Ω) for signal protection.
Pull-ups: Use 10kΩ on CS if required.



ESP32 DEV

I2C pull-ups: 4.7kΩ (SDA/SCL).
Series protection: 330Ω (GPIOs).
ADC calibration: Required for accuracy.
Reset: Add push-button across EN and GND.
Grounding: Shield analog signal lines.
Power stability: ripple <50mV.

GPIO voltage: 3.3V (max) use level shifters for >3.3V

CL/HDBME/17/01

ICBT CAMPUS

Sheet: /remote/

File: remote_controller.kicad_sch

Title: REMOTE_CONTROLLER

Size: A4

Date: 2024-12-30

Rev: PrageethM

KiCad E.D.A. kicad 7.0.9

Id: 3/3

